

Practice Set — for Francisco

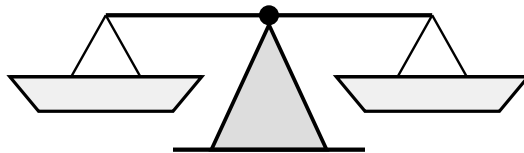
A few puzzles. Take your time, show your work where it helps.

Name _____

Date _____

1. Using the digits **4, 7, 2, 9** each exactly once, write the largest 4-digit number you can make. Then write the smallest.

Largest: _____ Smallest: _____



2. When you put coins on a balance scale, three different things can happen depending on what's on each side. List all three.

(a) _____

(b) _____

(c) _____

3. You have **2 coins**, and one of them is slightly heavier than the other. Using a balance scale, how can you find the heavy one in just **1 weighing**?

4. Now you have **3 coins**, and one is slightly heavier than the others. Find the heavy coin in just **1 weighing**. Describe how.

Hint: you don't have to put every coin on the scale.

5. You have **9 coins**, and one is slightly heavier than the rest. Find the heavy coin in just **2 weighings**. Describe your strategy.

Hint: think about what you just figured out in problem 4. Can you do it twice?

6. How many 2-digit numbers contain **no 7** anywhere in them?

Answer Key — for parents

- 1. Largest 9742, smallest 2479.** Biggest digit in the highest place for the largest; smallest digit first for the smallest.
- 2. The scale tilts left, tilts right, or balances.** Three outcomes — this is the key idea behind the rest of the coin problems. Each weighing gives you one of three answers, so each weighing can split a pile into three.
- 3. Put one coin on each side.** The heavy one is on the side that goes down. (Confidence builder.)
- 4. Weigh 1 coin against 1 coin, leave the third aside.** If the scale balances, the heavy coin is the one you set aside. If it tilts, the heavy coin is on the lower side. One weighing handles 3 coins because the scale has 3 outcomes (problem 2).
- 5. Split into 3 groups of 3.** *Weighing 1:* weigh group A against group B. If they balance, the heavy coin is in C; otherwise it's in the heavier of A or B. Now you have 3 suspect coins. *Weighing 2:* use the trick from problem 4 to find the heavy coin among those 3. Same trick, applied twice. With n weighings you can handle up to 3^n coins.
- 6. 72.** Tens digit can be 1–9 except 7 — 8 choices. Units digit can be 0–9 except 7 — 9 choices. $8 \times 9 = 72$. (Or: 90 two-digit numbers in total, minus 18 that contain a 7.)